Mobility-Aware UAV Positioning and Secret Key Rate Optimization for HAP–UAV CV-QKD

TBD*, TBD†
*XXX
†Computer Communications Laboratory, The University of Aizu, Aizuwakamatsu 965-8580, Japan

***Abstract*—This provisional internal-review study examines source selection and UAV positioning for a cloud-affected HAP–UAV discrete-modulated CV-QKD model. A finite catalog of 233 exact-source ensembles defines a common location-dependent numerical rate map. An offline stationary benchmark, a current-slot Greedy policy, and finite-horizon dynamic programming (DP) are compared under identical cloud realizations and speed constraints. Across 100 paired missions, expanding the candidate set from 10 to 100 raises the mean DP penalized objective by 11.41%. At blockage probability 0.8, mean zero-rate unavailability is 24.60%, 16.53%, and 4.07% for Stationary, Greedy, and DP, respectively. At the representative penalty, DP reduces travel by 88.37% relative to zero penalty while reducing mean usable rate by 0.515%. These are conditional manuscript-model outputs: the recorded two-dimensional mutual information and homodyne Holevo expression do not specify a consistent Bob measurement. The results characterize the saved numerical optimization and do not establish physical quantum security.**

***Index Terms*—Continuous-variable quantum key distribution, UAV, high-altitude platform, cloud blockage, probabilistic shaping, dynamic programming.**

Internal review only. Security-model correction and author review remain pending. Status:
`UNVALIDATED_SECURITY_MODEL`;
`NOT_READY_FOR_PUBLICATION_SCALE_RUNS`.

I. HAP–UAV CV-QKD System and Problem Formulation

A. Mobility-Aware UAV Positioning and SKR Optimization

We consider one UAV at a fixed altitude over a $1\,\mathrm{km}\times 1\,\mathrm{km}$ area, discretized into J candidate positions $\mathcal{Q}=\{\mathbf{q}_j\}_{j=1}^{J}$. The mission has N_t slots of duration Δt. Let $c_{j,n}\in\{0,1\}$ indicate cloud blockage (one means blocked). The matrix $\mathbf{C}=[c_{j,n}]$ contains spatially and temporally correlated Bernoulli states. P2 and P3 know the complete matrix and are offline benchmarks; Greedy observes only the current slot.

For an unblocked link, the recorded manuscript-style raw rate is

$$K_j(\nu,V_A)=\beta_{\mathrm{rec}}I_{AB,j}^{2D}(\nu,V_A)-\chi_{BE,j}(\nu,V_A). \tag{1}$$

Equation (1) is retained as a conditional numerical metric, subject to the measurement-consistency limitations below.

B. Recorded Physical-Layer Model and Security Boundary

Alice sends the coherent-state ensemble $\{p_k,\alpha_k\}_{k=1}^{256}$. With $[\hat{x},\hat{p}]=2i$ and vacuum quadrature variance one, the normalization is $V_A=2\sum_k p_k|\alpha_k|^2=2\bar{n}$. The source moments C_s,w_s are computed from this same ensemble; the subscript distinguishes them from the cloud matrix. For clear-sky power transmittance η, the saved evaluator uses

$$\begin{aligned}\eta_f&=\langle\sqrt{\eta}\rangle^2,\\ N=\eta_f\xi_f&=(\langle\eta\rangle-\eta_f)V_A+\xi_{\mathrm{ch}}\langle\eta\rangle,\\ V&=1+V_A,\qquad W=1+\eta_f V_A+N,\\ Z&=2\sqrt{\eta_f}C_s-\sqrt{2Nw_s}.\end{aligned}$$

The convention is $\xi_n=\xi_{\mathrm{ch}}$. The additive N is output-referred excess noise; loss-induced vacuum is already included in W's unit baseline. The active detector is ideal (efficiency one, electronic noise zero). No imperfect trusted-detector claim is made. The clear-sky distribution retains a separate angle-of-arrival outage atom at $\eta=0$ and its active density on $[0,1]$, without renormalizing. Moment reduction at the covariance level has a precedent for Gaussian modulation [1]; it is not an identity for the full discrete-input fading likelihood.

Specifically, the saved MI kernel uses the normalized likelihood

$$p_{\mathrm{num}}(\mathbf{y}|k)=\frac{\exp[-\|\mathbf{y}-2\sqrt{\eta_f}(\Re\alpha_k,\Im\alpha_k)\|^2/(2(1+N))]}{2\pi(1+N)}$$

and integrates both real components to obtain I_{AB}^{2D}. The Holevo branch uses the standard-form covariance with entries V,W,Z and the *homodyne* conditional eigenvalue

$$\nu_3^{\mathrm{hom}}=\sqrt{V(V-Z^2/W)}.$$

These two branches do not specify one consistent single-mode Bob measurement: a homodyne outcome has one quadrature, while joint measurement of both quadratures adds measurement vacuum. The measurement-dependent Holevo expressions are explicitly different in [2]. Consequently, all reported "usable SKR" values are conditional outputs of the recorded manuscript model, not established achievable secret-key rates.

Further, evaluating only Z_- does not establish the worst-case Holevo value over all admissible correlations; the general interval-maximization caveat in [2, Sec. 4, footnotes 11–12] also applies. A consistent measurement/noise specification, admissible correlation treatment, parameter estimation, and fading/outage conditioning protocol require author review. No correction is applied to the historical numerical results in this manuscript revision.

II. Joint Source and Mobility Optimization

Cloud blockage favors spatial adaptation, but UAV motion incurs travel cost. We jointly consider source selection, stationary deployment, and mobility-aware positioning. The finite catalog $\mathcal{G} = \{(\nu, V_A)\}$ contains 197 MB-shaped and 36 uniform 256-QAM ensembles. Source selection is

$$\mathcal{P}_1: \quad K_j^\star = \left[\max_{(\nu,V_A)\in\mathcal{G}} K_j(\nu, V_A)\right]^+. \tag{2}$$

Each candidate uses its own exact source moments and fading noise. Maximize raw rate before clipping; optimality is finite-catalog. With $S_{j,n} = (1 - c_{j,n})K_j^\star$, the best fixed position is

$$\mathcal{P}_2: \quad j_{\text{fix}}^\star = \arg\max_j \frac{1}{N_t}\sum_{n=1}^{N_t} S_{j,n}. \tag{3}$$

For a trajectory $j_n \in \{1, \ldots, J\}$, define $d_{i,j} = \|\mathbf{q}_i - \mathbf{q}_j\|_2$. The mobility problem is

$$\begin{aligned}\mathcal{P}_3: \quad \max_{\{j_n\}} \quad & \frac{1}{N_t}\sum_{n=1}^{N_t} S_{j_n,n} - \frac{\lambda}{D_{\text{ref}}}\sum_{n=2}^{N_t} d_{j_{n-1},j_n} \\ \text{s.t.} \quad & d_{j_{n-1},j_n} \le v_{\max}\Delta t, \quad n = 2, \ldots, N_t.\end{aligned} \tag{4}$$

Here $U = \overline{K} - \lambda D/D_{\text{ref}}$ balances mean usable rate $\overline{K}$ and total distance D; $\lambda \ge 0$ is in bit/use and $D_{\text{ref}} > 0$ normalizes distance. Staying and free initial placement are allowed, with no deployment or return cost. Distance is a cost surrogate, not calibrated propulsion energy.

III. Proposed Solution

P1 exhausts the catalog once per clear-sky position; P2 compares all J mission means. Let $\mathcal{N}(j) = \{i : d_{i,j} \le v_{\max}\Delta t\}$, including staying. Greedy starts at $j_1^{\text{G}} \in \arg\max_j S_{j,1}$ and then selects

$$j_n^{\text{G}} \in \arg\max_{j\in\mathcal{N}(j_{n-1}^{\text{G}})} \left[S_{j,n} - \frac{N_t\lambda}{D_{\text{ref}}} d_{j_{n-1}^{\text{G}},j}\right]. \tag{5}$$

This $O(N_t J)$ policy observes the current slot only. DP instead retains the best prefix ending at each position:

$$F_1(j) = S_{j,1}, \tag{6}$$

$$F_n(j) = S_{j,n} + \max_{i\in\mathcal{N}(j)} \left[F_{n-1}(i) - \frac{N_t\lambda}{D_{\text{ref}}} d_{i,j}\right]. \tag{7}$$

Store maximizing predecessors, choose the best terminal state, and backtrack. Then $U^\star = \max_j F_{N_t}(j)/N_t$ in dense worst-case time $O(N_t J^2)$, with $O(N_t J)$ backpointers and an $O(J^2)$ distance matrix. P2 and DP know the complete cloud sequence offline. DP has no smaller U than P2 or Greedy, but need not dominate unpenalized rate or availability. Its advantage also includes future information and initial-position selection.

TABLE I
Synthetic Baseline and Recorded Optical Parameters

Parameter	Value
HAP / UAV altitude	20 / 1 km
HAP horizontal position	$(11419.6551, 450)$ m
Flight region / candidates	1×1 km^2 / 100
Mission slots / slot duration	30 / 20 s
$v_{\max}$ / D_{ref}	15 m/s / 1000 m
Cloud (p, ℓ, ρ)	$(0.4, 250\,\text{m}, 0.9)$
Paired missions / seed	100 / 20261015
Exact source catalog	197 MB + 36 Uniform
Wavelength / full divergence	1550 nm / 233 μrad
Receiver diameter / τ_{zen}	0.08 m / 0.92
β_{rec} / ξ_{ch}	0.95 / 0.0186 SNU
Detector efficiency / noise	1 / 0
λ_{ref} (bit/use)	$9.468026677248516 \times 10^{-6}$

IV. Numerical Results and Discussion

Table I gives the synthetic baseline; cell centers are $50, 150, \ldots, 950$ m per axis. Gaussian-copula clouds use spatial covariance $\exp(-d_{i,j}/\ell)$ and latent AR(1) coefficient ρ. Comparisons share 100 mission IDs (0–99); uncertainty uses saved whole-mission bootstrap intervals. Availability means $S > 0$, without an engineering threshold. The presentation follows [3].

Fig. 1 compares the positioning schemes as blockage increases. Observed usable rates decrease, but movement can avoid blocked locations. At $p = 0.8$, mean zero-rate unavailability is 24.60%, 16.53%, and 4.07% for Stationary, Greedy, and DP, respectively. This availability benefit is observed; P3 optimizes U, not positive-rate slot count.

Fig. 2 shows persistent favorable locations making offline stationary selection competitive. As ρ increases from 0 to 0.98, mean DP distance falls from 1038.25 to 31.84 m, while its mean objective gain over P2 falls from 23.712% to 0.155%. This sampled trend is not a correlation monotonicity theorem or an online result. Enlarging nested position sets, however, cannot lower optimal P2/DP U: increasing J from 10 to 100 improves mean DP U by 11.409%.

At the default configuration and λ_{ref}, DP improves mean usable rate by 4.16% over P2 and 2.70% over Greedy, and mean objective by 3.07% and 4.42%, respectively. Relative to zero-penalty DP, the representative penalty reduces mean travel by 88.370% with only a 0.515% decrease in mean usable rate. All locations select the same MB source $(\nu, V_A) = (0.125, 8)$ at the catalog boundary; these gains do not isolate shaping gain. Rates retain the security limitations of Section I-B; finite-key, tracking, and nonideal-detector effects [4] are excluded.

V. Conclusion

Finite-catalog MB source selection and cloud-aware positioning were compared through Stationary, current-slot Greedy, and offline DP. DP optimizes the finite-grid penalized objective, with observed blockage avoidance and a favorable rate–distance tradeoff. Findings remain conditional on synthetic clouds, offline information, and a distance-cost surrogate. The two-dimensional MI and homodyne-conditioned

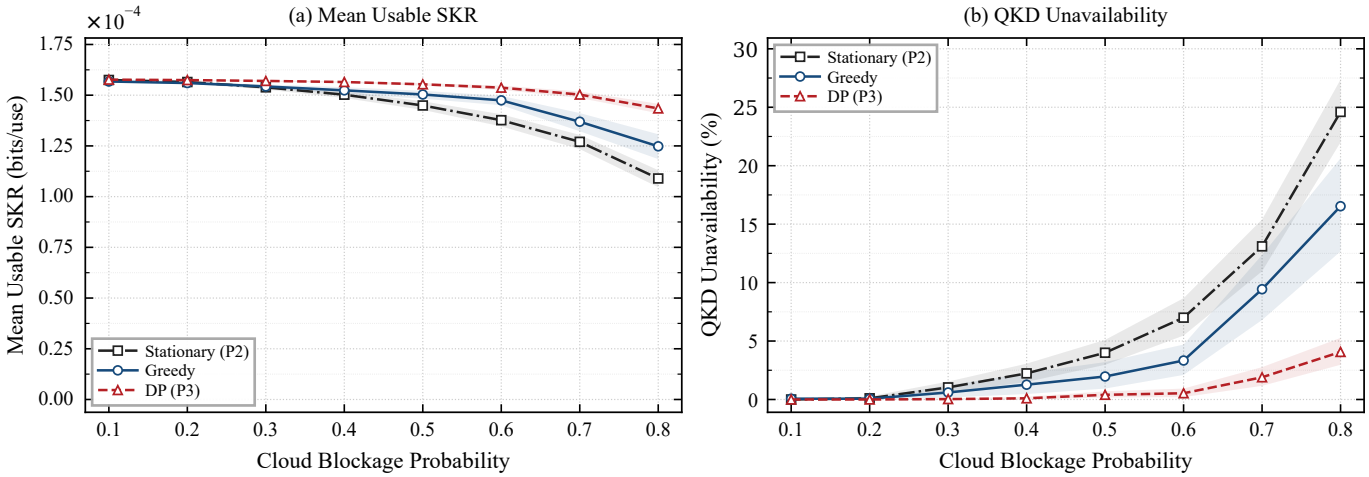


Fig. 1. Cloud blockage sensitivity at λ_{ref}: (a) mean usable rate; (b) zero-rate unavailability. Stationary, Greedy, and DP share 100 paired missions; bands are pointwise 95% whole-mission bootstrap intervals.

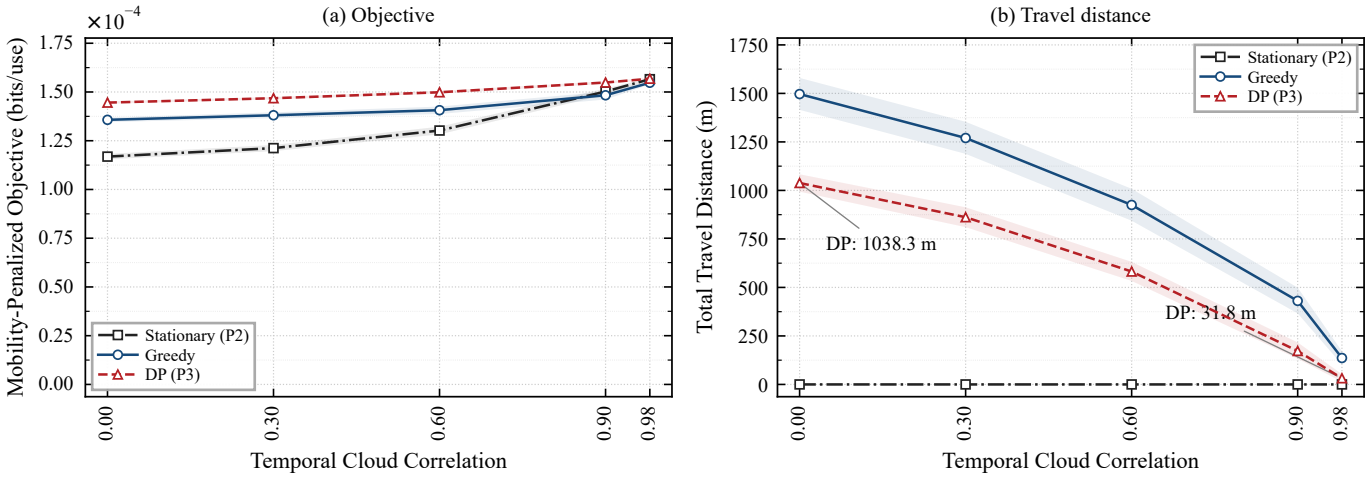


Fig. 2. Latent temporal correlation at $p = 0.4$, $\ell = 250$ m, and λ_{ref}: (a) mean objective; (b) total travel distance. Methods share 100 mission IDs; bands are pointwise 95% whole-mission bootstrap intervals. Latent ρ differs from binary cloud correlation.

Holevo expressions are not measurement-consistent; a consistent security model is required before these numerical rates can represent secure keys.

References

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